

Mohana Uma Sai Manem

Boston, MA | Open to Relocate

508-304-4616 | manemmohansaicareers@gmail.com | LinkedIn: <https://www.linkedin.com/in/mohanamanem>

GitHub: <https://github.com/manemsai>

Summary

Results-driven Data Scientist & AI Engineer with over 9 years of experience in machine learning, AI, data science, and business intelligence, specializing in predictive analytics, NLP, deep learning, and cloud-based architectures (AWS, Azure). Proficient in ML algorithms (Random Forests, SVM, XGBoost, Neural Networks), big data technologies (PySpark, Hadoop, Databricks), and MLOps frameworks (MLflow, DVC, Airflow, Docker, Kubernetes) to build and deploy scalable AI solutions. Skilled in data engineering, ETL, SQL/NoSQL databases (PostgreSQL, SQL Server, Snowflake) and data visualization (Power BI, Tableau, Matplotlib) to drive data-driven insights. Strong expertise in LLMs, GenAI applications, chatbot development, and AI optimization with hands-on experience in scalability, caching, rate-limiting, and low-latency API handling. Proven ability to lead cross-functional teams, optimize forecasting models, and deliver AI-powered business solutions across industries like energy, finance, and e-commerce.

Technical Skills

Neural Networks | Regression | Classification | Clustering | Transformers | LLM | Fine-tuning | Agentic AI | Retrieval-Augmented Generation (RAG) | NLP | Time Series Analysis | Deep Learning | Scikit-learn | PyTorch | TensorFlow | Pandas | NumPy | Matplotlib | Seaborn | Lang Chain for LLM orchestration | Data pipelines | Data lakes | Data warehouses | Python | SQL | PySpark | MATLAB | AWS (Sage Maker | Bedrock | Athena | Red Shift | Elastic Beanstalk | Glue | EC2) | Azure (ML, Data Factory) | AI Strategy | Cross-Functional Collaboration | Technical Mentorship | Stakeholder Communication.

Experience

ElevateMe, Columbus, Ohio

02/2025 - Present

Data Analytics & Machine Learning Fellow Trainee

- Engaging in 150+ hours of hands-on learning and project work, including active participation in a live project and two capstone projects.
- Utilize advanced tools such as Python, Jupyter Notebooks, Pandas, Numpy, Matplotlib, Seaborn, and Scikit-learn to build and evaluate machine learning models.
- Deploy machine learning models on Microsoft Azure, integrating Azure ML with Azure SQL databases to enable real-time analytics and predictions.
- Perform data preprocessing, feature engineering, and model selection, and evaluate regression, classification, and clustering algorithms to optimize performance.
- Collaborate in a professional team environment using tools like Jira, Confluence, and Slack for project management and effective communication.

Awsome LLC, San Diego, California

05/2024 – 12/2024

Machine Learning Intern, Talk to My Energy (TTME) Project

- Built an automated conversational AI agent using PandasAI, integrating OpenAI, Gemini, and AWS Bedrock models, reducing customer response time by 35% and improving customer satisfaction by 18%.
- Automated data retrieval for customer service representatives, enabling real-time access to customer billing and complaint data, reducing overhead expenses by 20%.
- Created detailed analytical reports on customer interactions and service performance using Power BI, improving transparency and decision-making for the management team.
- Streamlined workflows by integrating conversational AI with CRM systems, reducing dependency on manual intervention by 25%.
- Leveraged AWS Glue and Redshift to design and implement end-to-end ETL pipelines to retrain the model using customer feedback, reducing data preprocessing time by 25% and enabling real-time analytics

NLC Tamil Nadu Power Limited; Tuticorin, India

Project Manager – Data Engineering & Business Transformation

September 2020 – August 2023

- Reduced equipment downtime by 22% by leading the implementation of predictive maintenance systems using IoT data, optimizing energy distribution systems and cutting maintenance costs by 18%.
- Improved operational performance by 10% and team efficiency by 30% by integrating cross-departmental dashboards, fostering collaboration, and breaking down data silos.
- Automated financial reporting and contract tracking systems, reducing manual work by 35% and saving 10% on operational costs by improving financial oversight.
- Delivered projects 20% faster and reduced budget overruns by 15% by introducing Agile project management practices and ensuring timely deliveries through scrum methodologies.

Data Analyst – Energy Analytics & Business Intelligence

September 2016– September 2020

- Achieved 89% accuracy in energy demand forecasting by developing and deploying machine learning models (XGBoost, Random Forest), optimizing energy production and reducing unnecessary generation costs by 12%.
- Built and maintained real-time Power BI dashboards, improving energy efficiency by 15% by providing actionable insights into KPIs and energy consumption trends.
- Reduced peak demand by 18% by applying Gaussian Mixture Models to segment customers based on energy usage and implementing targeted energy-saving campaigns.
- Improved data consistency by 30% by developing data validation systems, ensuring data integrity and enabling more reliable analysis.

Executive Trainee – Engineering & Operations.

September 2014 – September 2016

- Reduced downtime by 15% by assisting in the monitoring and maintenance of electrical equipment (transformers, generators, substations), implementing preventive maintenance schedules to improve system reliability and minimizing failures.
- Increased energy production efficiency by 5% by analyzing performance data and recommending system optimizations.
- Improved regulatory compliance by assisting in the documentation and reporting of system performance and maintenance activities, leading to 20% faster report submission.
- Participated in troubleshooting efforts, resolving equipment issues 15% faster by leveraging predictive maintenance techniques.
- Developed an asset management system for tracking equipment health, improving 8% equipment utilization through better management of underperforming assets.

Education

Clark University, Worcester, Massachusetts

08/2023-12/2024

Master of Science in Data Analytics (CGPA: 3.96/4.0)

Relevant Course Work: Data Modeling, Business Intelligence, Linear Regression, Machine Learning, Deep Learning, Natural Language Processing.

Jawaharlal Nehru Technological University, Kakinada, India

09/2009-05/2013

Bachelor of Technology in Electrical and Electronics Engineering (CGPA 3.5/4.0)

CERTIFICATIONS

Data Analytics and Machine Learning Certificate of Completion

02/2025-Present

ElevateMe

150+ hours of hands-on coursework, having completed a Data Analytics project and Machine Learning project using Tableau, Python, Machine Learning and SQL

Data Analytics with Python

10/2023-10/2023

Coursera

Understand the Python for Data Analysis (Pandas, NumPy), Data Visualization (Matplotlib, Seaborn), Machine Learning Basics (Scikit-learn), Statistical Data Analysis concepts.